

Bharat O2

Bharat O2 is also an open source oxygen concentrator designed in India that can be made using locally available parts. This design does not use any solenoid valve or electrical relay, which reduces the chances of potential failure. Instead, a custom-built PSA valve is used that reduces the build cost as well. The valve has been made using motor parts by the design team. Bharat O2 has a 4-way swing valve that eliminates the use of conventional parts used in the Pressure Swing



Adsorption technique, hence eliminating sixteen mechanical and six electrical failure points. It can deliver oxygen with a concentration of 93 % at five litres per minute.

For details: <https://bharato2.org>

Open OX

Open OX delivers oxygen with a concentration of 85 to 95 % at flow rate of six litres per minute and consumes 580 watts of power. It uses Zeolite Adsorption technology for oxygen separation. The motherboard of this oxygen concentrator provides a connector that is suitable for most medical-grade oxygen sensors. While the electro-galvanic sensors have a low life span, it gives you the flexibility to use ultrasonic oxygen sensors that use the speed of sound to measure the amount of oxygen in a gas or liquid sample.



For details: <https://openox.hacklab.kiev.ua>

OxiKit



OxiKit can deliver oxygen concentration of 92 % with a flow rate of 24 litres per minute while most other oxygen concentrators are able to deliver only 5 to 15 litres per minute. The Azure IoT Hub is developing the design with more features like Wi-Fi connectivity. Device data from the pulse oximeter and concentrator can be automatically uploaded to the Global Telehealth Services. The flow rate can also be automated based on the oxygen saturation level and Bluetooth pulse oximeter readings.

For details: <https://oxikit.com/>

OX

Developed by PubInv, OX open source oxygen concentrator aims at providing 95 % medical-grade oxygen concentration with a flow rate of ten litres per minute, which is good considering the flow rate.

For details: <https://github.com/PubInv/Ox>
<https://app.jogl.io/project/580?t=documents>



Conclusion

Overall, the most efficient and innovative open source design among the above-mentioned is Bharat O2. It eliminates sixteen mechanical and six electrical points and thus reduces chance of failure. Also, the custom-built valve helps reduce the cost of the project.

If we need a high flow rate with good oxygen concentration level, then OxiKit and MARUT may be preferred as these are able to deliver a good flow rate—ten litres per minute in case of Marut and twenty-four litres per minute in case of OxiKit—without compromising the oxygen concentration level. **EFY**